

# Leakage-safe forecasting of German grid load

Daily and fixed 24-hour-ahead forecasts with reconciliation and conformal intervals

## Abstract

This study asks how accurately German electricity demand can be forecast from a restricted forecast-origin information set. We compare one-day-ahead daily models, fixed 24-hour-ahead hourly forecasts, daily-hourly reconciliation, and three forms of uncertainty quantification. Model choice is fixed on a chronological validation period and a separate calibration period is used for interval construction. On the previously inspected final period, the daily ensemble achieved 1.855% MAPE and 0.442 MASE. Reconciliation reduced hourly MAE from 1624.9 MW to 1562.1 MW. A paired day-block bootstrap did not distinguish the residual hybrid from direct LightGBM. Adaptive 90% intervals reached 89.95% empirical coverage, but this retrospective result does not provide a prospective guarantee.

# 1 Introduction

Load forecasts inform scheduling and operational planning, yet a retrospective model can appear stronger than it is if it uses weather observations that were unavailable when the forecast would have been issued. This study treats forecast origin as part of the prediction problem. The analysis asks two related questions: how accurately the next daily mean can be predicted, and how accurately each valid hour can be predicted exactly 24 hours earlier.

The analysis compares daily statistical and tree models, three direct hourly designs, a constraint that aligns hourly profiles with an independently forecast daily mean, and three interval methods. The dashboard offers a compact tour of the evidence. This report develops the experimental assumptions, statistical comparisons, error structure, and limitations in greater depth.

## Release results

- Period: 01 Jan 2026 to 04 Aug 2026;  $n=216$  days.
- Daily forecast: Ensemble. MAE 997.7 MW, MAPE 1.855%, and MASE 0.442. Reference models: Naive (t-1) MAPE 7.025% ( $n=216$ ); Seasonal naive (t-7) MAPE 5.575% ( $n=216$ ).
- Hourly forecast: Residual hybrid. Daily-total alignment reduced MAE from 1624.9 MW to 1562.1 MW (-3.86%). The result contains  $n=5183$  hourly values. The daily model was LightGBM.
- Model comparison: Validation MAE was 1463.6 MW for Residual hybrid and 1466.6 MW for Direct LightGBM. The paired difference on the final data was +1.6 MW. Its 95% range was [-44.5, 46.7] MW across 216 days. This range includes zero, so the data do not show a clear winner.
- Uncertainty ranges: Symmetric 90%: target 90%, measured coverage 85.18%, mean width 5612.0 MW, interval score 8837.3 MW,  $n=5183$ ; Adaptive 90%: target 90%, measured coverage 89.95%, mean width 6545.0 MW, interval score 8273.5 MW,  $n=5183$ ; CQR 90%: target 90%, measured coverage 86.67%, mean width 7556.4 MW, interval score 10519.8 MW,  $n=5183$ .
- Scope: The final period was already examined. These results describe this data period and do not state future accuracy.

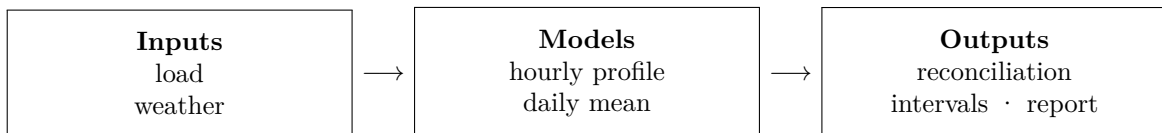
## 2 Data and evaluation design

The target is realized German electricity consumption reported by SMARD (filter 410), expressed in MW. Hourly observations are normalized in UTC and mapped to the Europe/Berlin local calendar. Open-Meteo supplies temperature at 2 m and wind speed at 10 m for Berlin, Hamburg, Munich, Cologne, and Frankfurt. Population weights combine these series into a national weather proxy. It captures broad exposure rather than regional grid heterogeneity.

The dataset contains 2,760 daily observations from 14 January 2019 through 4 August 2026. From 20 January 2024 onward, weather features use archived forecasts issued 24 hours before the target day. Earlier dates use the previous day’s realized weather because an equivalent forecast archive is unavailable. The two ranges therefore differ in weather-information quality.

The chronological split is training through 2023, validation from January 2024 through June 2025, calibration from July through December 2025, and final evaluation from January through 4 August 2026. Model and feature decisions use validation evidence only. Calibration outcomes set interval widths but do not select the point model. The final period contains 216 local days and 5183 hourly observations. Because that period was inspected during development, it is reported as retrospective evidence rather than an untouched holdout.

## 3 Forecasting methods



Daily predictors include one- and seven-day load lags, weekly and annual Fourier terms, holidays, weekend and bridge-day indicators, and heating and cooling degree variables. Candidate forecasts comprise SARIMAX, XGBoost, LightGBM, Random Forest, two persistence baselines, and an ensemble of SARIMAX, XGBoost, and LightGBM fixed from validation evidence.

For observations  $y_t$  and forecasts  $\hat{y}_t$ , the main measures are

$$\text{MAE} = n^{-1} \sum_{t=1}^n |y_t - \hat{y}_t|, \quad \text{MAPE} = 100n^{-1} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right|.$$

MASE scales absolute error by the in-sample one-step naive error; values below one improve on that scale reference. RMSE is retained to reveal sensitivity to large errors.

The hourly evaluation compares residual hybrid, direct LightGBM, and direct Ridge. All candidates predict valid horizons directly and use horizon, 24- and 168-hour lags, weather, calendar, and Fourier features. Direct Ridge provides a linear reference. Direct LightGBM fits the nonlinear relationship in one model. The hybrid first estimates a linear trend and then fits LightGBM to its residuals. Direct prediction avoids recursive error accumulation. Reconciled validation MAE selects the hourly model; later rankings do not revise that decision. The released hourly metrics use only horizon 24, so each row has its own origin at valid time minus 24 hours. They are not a common-origin profile issued once for the entire next local day.

## 4 Reconciliation

For local day  $d$  and hour  $h$ , the system uses:

$$\hat{y}_{d,h}^{rec} = \hat{y}_{d,h}^{hourly} \times \frac{\hat{y}_d^{daily}}{\frac{1}{|H_d|} \sum_{j \in H_d} \hat{y}_{d,j}^{hourly}}.$$

The operation keeps the relative hourly shape. The reconciled mean equals the daily forecast.

The code supports complete local days with 23, 24, or 25 hours. It rejects invalid predictions and incomplete days.

Raw and reconciled MAE use the same forecast rows, so the difference isolates the daily constraint rather than a change in sample. Reconciliation is performed before conformal scores are calculated.

## 5 Model selection

The release keeps the validation-selected model. Later candidate comparisons are descriptive and do not change that selection.

The paired bootstrap resamples complete local days, preserving dependence among hourly errors within a day. Its estimand is candidate minus reference MAE. A 95% interval containing zero does not establish a clear difference between the models.

## 6 Prediction intervals

The point, daily, and quantile models use data before calibration. The models stay fixed during calibration and final evaluation. Symmetric split conformal intervals use frozen calibration residuals. The adaptive variant updates its level sequentially with  $\gamma = 0.01$  and bounded history. CQR estimates conditional lower and upper quantiles before conformal calibration.

Reconciliation occurs before score calculation. Both interval methods use the same reconciled point forecasts. Coverage, mean width, and interval score are considered together because coverage alone can be increased by producing uninformatively wide ranges. Adaptive coverage is prequential monitoring evidence and does not constitute an unconditional time-series guarantee under serial dependence.

## 7 Results

### 7.1 Daily forecasts

Table 1: Daily performance on the final evaluation period.

| Model                | RMSE (MW) | MAE (MW) | MAPE (%) | MASE  |
|----------------------|-----------|----------|----------|-------|
| SARIMAX              | 1465.6    | 1090.6   | 2.040    | 0.484 |
| XGBoost              | 1371.6    | 1074.2   | 1.999    | 0.476 |
| LightGBM             | 1413.8    | 1127.8   | 2.095    | 0.500 |
| Random Forest        | 1567.4    | 1154.0   | 2.139    | 0.512 |
| Naive (t-1)          | 5160.6    | 3658.1   | 7.025    | 1.622 |
| Seasonal naive (t-7) | 4189.9    | 2978.8   | 5.575    | 1.321 |
| Ensemble             | 1302.8    | 997.7    | 1.855    | 0.442 |

The ensemble produced the lowest final-period error: MAE 997.7 MW, MAPE 1.855%, and MASE 0.442.

It substantially outperformed both naive baselines and also led validation MAPE, so the reported choice is consistent with the selection period rather than being chosen from the final ranking. Its final-period result is reassuring for this sample but does not establish stability beyond August 2026.

## 7.2 Hourly forecasts and reconciliation

The residual hybrid had reconciled validation MAE 1463.6 MW, compared with 1466.6 MW for direct LightGBM, and was fixed as the selected hourly model. On the final period, reconciliation reduced its MAE from 1624.9 MW to 1562.1 MW, an improvement of 3.86%.

Direct LightGBM recorded final reconciled MAE 1558.0 MW, slightly below the selected hybrid’s 1562.1 MW. The paired day-block estimate of their MAE difference was 1.6 MW, with a 95% interval of  $[-44.5, 46.7]$  MW across 216 days. The estimated probability that the candidate was better was 47.74%. The evidence supports practical equivalence at the resolution of this experiment, not a post-hoc model switch.

## 7.3 Prediction intervals

Table 2: Hourly 90% interval performance on 5183 observations.

| Method    | Coverage | Mean width (MW) | Interval score (MW) |
|-----------|----------|-----------------|---------------------|
| Symmetric | 85.18%   | 5612.0          | 8837.3              |
| Adaptive  | 89.95%   | 6545.0          | 8273.5              |
| CQR       | 86.67%   | 7556.4          | 10519.8             |

The fixed symmetric interval under-covered at 85.18%. CQR was wider but still under-covered at 86.67%. The adaptive method reached 89.95% and achieved the lowest interval score, although its mean width rose to 6545.0 MW. Adaptation improved the observed calibration–sharpness trade-off in this period without supplying a future guarantee.

## 7.4 Weather contribution and error structure

Replacing forecast-origin weather with a previous-day persistence proxy modestly increased ensemble MAPE. Weather adds measurable value, but the small change shows that recent and seasonal load structure carries most of the signal at this national one-day horizon. Errors are not assumed independent across hours: model comparison uses day blocks, while interval and drift diagnostics retain monthly and horizon slices.

# 8 Data validation

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| <b>Load</b>       | Values must be finite and positive.                              |
| <b>Weather</b>    | Values must be within configured physical ranges.                |
| <b>Time</b>       | UTC timestamps must be unique, complete, and on the hourly grid. |
| <b>Evaluation</b> | Prediction and interval indexes must match before the join.      |

# 9 Discussion

Three conclusions follow from the experiment. A modest ensemble supplies an accurate daily anchor without adding complexity after validation. The daily constraint improves the selected hourly forecast while preserving its intraday shape. Finally, two hourly tree-based designs are too close to justify claiming that either is generally superior. This negative result favors retaining the validation choice instead of tuning against inspected outcomes.

The weather ablation qualifies the role of exogenous inputs. Forecast-origin weather improves error, but lagged demand and calendar structure dominate at this scale. More spatial detail may matter during regional extremes, yet testing that proposition requires regional load targets or a separate aggregation study rather than simply adding stations.

# 10 Limitations

The evaluation period was inspected during development, so its numerical results describe this release and are not prospective accuracy claims. The pre-2024 weather fallback is weaker than an archived issue-time forecast. Five population-weighted cities cannot represent every regional weather or industrial-load effect. SMARD revisions, market changes, and distribution shift can alter point error and interval coverage.

The hourly headline is a fixed 24-hour-ahead evaluation, not a market-style profile issued once for all hours of the next local day. The study concerns offline forecasting. It does not test data-arrival latency, scheduler reliability, access control, or service-level objectives. Those are deployment questions and should not be inferred from a retrospective model evaluation.

## 11 Reproducibility

The repository fixes split dates, feature definitions, parameters, and random seed in `config.yaml`. A clean run downloads public inputs, rebuilds features, evaluates the frozen protocols, compiles this report, validates artifact hashes, and emits the bounded dashboard payload. Detailed CSV tables are generated on demand by the manual GitHub Actions workflow rather than stored in Git. Source, protocol, and artifact hashes remain available in `results/run_summary.json` and `results/evaluation_protocol.json`; they are omitted from the scientific narrative because they support verification rather than interpretation.

Tree-model parameters are frozen values inherited from earlier experimentation. The complete historical search trials are not available, so the study is reproducible from the committed parameters onward but does not claim reproducibility of the original search.

## References

- Bundesnetzagentur. SMARD electricity-market data. <https://www.smard.de/en>
- Open-Meteo. Historical Weather and Previous Runs APIs. <https://open-meteo.com/en/docs/previous-runs-api>
- Hyndman, R. J. et al. “Optimal combination forecasts for hierarchical time series.” *Computational Statistics & Data Analysis*, 2011.
- Ke, G. et al. “LightGBM: A highly efficient gradient boosting decision tree.” *Advances in Neural Information Processing Systems*, 2017.
- Romano, Y. et al. “Conformalized quantile regression.” *Advances in Neural Information Processing Systems*, 2019.
- Gibbs, I. and Candès, E. “Adaptive conformal inference under distribution shift.” *Advances in Neural Information Processing Systems*, 2021.